

EXPERIMENT – 25

Case Scenario : MySQL Queries Implementation for QR Enabled Automatic Bus Ticket Booking System

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mysql>
mysql> -- EXPERIMENT 25: Case Scenario - QR Enabled Automatic Bus Ticket Booking System
mysql> DROP DATABASE IF EXISTS bus_db;
Query OK, 4 rows affected (0.06 sec)

mysql> CREATE DATABASE bus_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE bus_db;
Database changed
mysql>
mysql> CREATE TABLE Route (
  ->     route_no INT PRIMARY KEY,
  ->     source VARCHAR(30) NOT NULL,
  ->     destination VARCHAR(30) NOT NULL,
  ->     distance_km INT
  -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE Bus (
  ->     bus_no INT PRIMARY KEY,
  ->     bus_name VARCHAR(50) NOT NULL,
  ->     route_no INT,
  ->     capacity INT NOT NULL,
  ->     FOREIGN KEY (route_no) REFERENCES Route(route_no)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE Passenger (
  ->     passenger_id INT PRIMARY KEY,
  ->     name VARCHAR(50) NOT NULL,
  ->     phone VARCHAR(15)
  -> );
Query OK, 0 rows affected (0.02 sec)
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mysql>
mysql> CREATE TABLE Ticket (
->   ticket_id INT AUTO_INCREMENT PRIMARY KEY,
->   passenger_id INT,
->   bus_no INT,
->   travel_date DATE NOT NULL,
->   qr_code VARCHAR(100) UNIQUE,
->   fare DECIMAL(6,2),
->   FOREIGN KEY (passenger_id) REFERENCES Passenger(passenger_id),
->   FOREIGN KEY (bus_no) REFERENCES Bus(bus_no)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> INSERT INTO Route VALUES
-> (1,'Chennai','Vellore',140),
-> (2,'Chennai','Trichy',320),
-> (3,'Chennai','Pondicherry',160);
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO Bus VALUES
-> (101,'TN Express 1',1,50),
-> (102,'TN Express 2',2,45),
-> (103,'TN Express 3',3,50);
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql>
mysql> INSERT INTO Passenger VALUES
-> (1,'Karthik R','9880011111'),
-> (2,'Meena S','9880022222'),
-> (3,'Rahul Dev','9880033333');
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0

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mysql>
mysql> INSERT INTO Ticket (passenger_id, bus_no, travel_date, qr_code, fare) VALUES
-> (1,101,'2026-08-05','QR-1-101-20260805',150.00),
-> (2,102,'2026-08-06','QR-2-102-20260806',320.00),
-> (3,103,'2026-08-05','QR-3-103-20260805',180.00);
Query OK, 3 rows affected (0.00 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql>
mysql> SELECT t.qr_code, p.name, b.bus_name, r.source, r.destination, t.travel_date, t.fare
-> FROM Ticket t
-> JOIN Passenger p ON t.passenger_id = p.passenger_id
-> JOIN Bus b ON t.bus_no = b.bus_no
-> JOIN Route r ON b.route_no = r.route_no;
+-----+-----+-----+-----+-----+-----+-----+
| qr_code | name | bus_name | source | destination | travel_date | fare |
+-----+-----+-----+-----+-----+-----+-----+
| QR-1-101-20260805 | Karthik R | TN Express 1 | Chennai | Vellore | 2026-08-05 | 150.00 |
| QR-2-102-20260806 | Meena S | TN Express 2 | Chennai | Trichy | 2026-08-06 | 320.00 |
| QR-3-103-20260805 | Rahul Dev | TN Express 3 | Chennai | Pondicherry | 2026-08-05 | 180.00 |
+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
mysql> SELECT b.bus_name, COUNT(*) AS tickets_booked
-> FROM Ticket t JOIN Bus b ON t.bus_no = b.bus_no
-> GROUP BY b.bus_name;
+-----+-----+
| bus_name | tickets_booked |
+-----+-----+
| TN Express 1 | 1 |
| TN Express 2 | 1 |
| TN Express 3 | 1 |
+-----+-----+
3 rows in set (0.00 sec)

```

```
mysql>
mysql> SELECT r.source, r.destination, SUM(t.fare) AS total_revenue
-> FROM Ticket t
-> JOIN Bus b ON t.bus_no = b.bus_no
-> JOIN Route r ON b.route_no = r.route_no
-> GROUP BY r.source, r.destination;
```

source	destination	total_revenue
Chennai	Vellore	150.00
Chennai	Trichy	320.00
Chennai	Pondicherry	180.00

3 rows in set (0.00 sec)

```
mysql>
mysql> SELECT * FROM Ticket WHERE qr_code = 'QR-1-101-20260805';
```

ticket_id	passenger_id	bus_no	travel_date	qr_code	fare
1	1	101	2026-08-05	QR-1-101-20260805	150.00

1 row in set (0.00 sec)